

Turtle Drawing Cheatsheet

A printable reference for the turtle drawing template. Edit `src/turtle_drawing/drawings.py`, save, and use `uv run interactive` for real-time drawing.

<code>uv run interactive</code>	Real Turtle window. Reloads quickly when you save.
<code>uv run live</code>	Auto-refresh PNG preview window.
<code>uv run render</code>	Generate output/drawing.png once.
<code>uv run watch</code>	Regenerate output/drawing.png whenever code changes.

Move

```
t.forward(100)
t.backward(50)
t.left(90)
t.right(45)
t.goto(10, 20)
t.setheading(0)
t.home()
```

Draw

```
t.circle(40)
t.dot(20, "green")
t.write("Hello")
t.clear()
t.hideturtle()
t.speed(1) # slow
t.speed(0) # fastest
```

Pen

```
t.penup()
t.pendown()
t.pensize(5)
t.color("red")
t.pencolor("blue")
t.fillcolor("yellow")
current = t.pencolor()
```

Fill

```
t.fillcolor("gold")
t.begin_fill()
draw_square(t, 80)
t.end_fill()
draw_filled_circle(t, 25, "red", "pink")
```

Coordinates

(0, 0) is the center
x: left / right
y: down / up
`jump_to(t, x, y)`

Run It

```
uv sync
uv run interactive
uv run live
uv run render
uv run watch
uv run draw
```

Most important idea

The turtle remembers where it is pointing. Movement commands draw from the current position. Use `t.penup()` or `jump_to(t, x, y)` when you want to move without drawing.

Drawing Recipes and Helper Functions

Use these helpers from **drawings.py** to build pictures faster. You can copy the recipes and change numbers or colors.

Shape helpers

<code>draw_square(t, 80)</code>	Square using the current pen color.
<code>draw_rectangle(t, 120, 60)</code>	Rectangle with width and height.
<code>draw_triangle(t, 90)</code>	Equilateral triangle.
<code>draw_circle(t, 35)</code>	Circle above the turtle.
<code>draw_star(t, 70)</code>	Five-point star.
<code>draw_flower(t, x, y)</code>	Flower centered near x, y.

Filled shape helpers

<code>draw_filled_square(t, 80, "black", "gold")</code>	Filled square.
<code>draw_filled_rectangle(t, 120, 60, "brown", "tan")</code>	Filled rectangle.
<code>draw_filled_triangle(t, 100, "red", "tomato")</code>	Filled triangle.
<code>draw_filled_circle(t, 30, "blue", "lightblue")</code>	Filled circle.

House Recipe

```
jump_to(t, -100, -50)
draw_filled_square(t, 160, "brown", "tan")
jump_to(t, -120, -50)
draw_filled_triangle(t, 200, "red", "tomato")
```

Repeating Pattern

```
for _ in range(12):
    draw_star(t, 50)
    t.right(30)
    t.forward(20)
```

Spiral

```
for size in range(10, 160, 10):
    t.forward(size)
    t.right(90)
```

Color Palette

```
"red", "orange", "gold"
"green", "limegreen"
"blue", "lightblue"
"purple", "plum"
"black", "gray", "white"
"brown", "tan", "pink"
```

Write Labels

```
jump_to(t, -80, 120)
t.color("black")
t.write("My picture")
t.write("Big", font=("Arial", 20, "bold"))
```

Reusable Function

```
def draw_tree(t, x, y):
    jump_to(t, x, y)
    draw_filled_rectangle(t, 30, 100, "brown", "peru")
    jump_to(t, x - 25, y + 75)
    draw_filled_circle(t, 45, "green", "limegreen")
```

Interactive, Images, and Debugging

`uv run interactive` reloads quickly after every save and remembers the window position. `uv run render` and `uv run live` create PNG previews.

Interactive hooks

<code>def on_click(t, x, y):</code>	Runs when the user clicks the Turtle window.
<code>def on_key(t, key):</code>	Runs when the user presses a supported key.
<code>r / g / b</code>	Change the current pen color in the example.
<code>arrow keys</code>	Move the turtle and draw in the example.
<code>space</code>	Draw a star in the example.
<code>t.clear()</code>	Clear the window.
<code>draw_picture(t)</code>	Draw the starting picture again.

Images and pixels

<code>import_picture(t, "assets/example_picture.png", 0, 0, width=120)</code>	Place an image in the PNG preview.
<code>pixel = get_pixel_color(t, 0, 0)</code>	Read an RGB color from the PNG preview.
<code>draw_color_swatch(t, -80, 20, pixel)</code>	Draw a square showing that RGB color.

Click With Current Color

```
def on_click(t, x, y):  
    current = t.pencolor()  
    jump_to(t, x, y - 20)  
    draw_filled_circle(t, 20, current, current)
```

Keyboard Example

```
def on_key(t, key):  
    if key == "r":  
        t.color("red")  
    if key == "space":  
        draw_star(t, 50)
```

Debug Checklist

Save `drawings.py`
Read the terminal error
Check parentheses: ()
Check quotes: "red"
Check indentation
Try `uv run render`

Common Mistakes

```
forward(100) # missing t.  
t.color(red) # missing quotes  
t.goto(10) # needs x and y  
if key = "r": # use ==
```

When to use each preview

Use `interactive` for real-time Turtle drawing and mouse/keyboard experiments. Use `live` or `render` when you need image import, pixel reading, or a saved PNG.